

Epinephrine (adrenaline) (systemic): Drug information - UpToDate

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For additional information see "Epinephrine (adrenaline) (systemic): Patient drug information" and "Epinephrine (adrenaline) (systemic): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Adrenalin;
- Adrenalin-NaCl;
- Auvi-Q;
- Epinephrine Professional;
- EPINEPHrinesnap;
- EpinephrineSnap-EMS;
- Epinephrinesnap-v;
- EpiPen 2-Pak;
- EpiPen Jr 2-Pak;
- EPlsnap [DSC];
- Neffy;
- Symjepi [DSC]

Brand Names: Canada

- Adrenalin;
- Allerject [DSC];
- Anapen Junior [DSC];
- Anapen [DSC];
- Emerade [DSC];
- EpiPen;
- EpiPen Jr;
- Neffy

Pharmacologic Category

- Alpha-/Beta- Agonist

Dosing: Adult

Dosage guidance:

Safety: Epinephrine injectable dosage form concentrations are no longer labeled as ratio expressions (eg, 1:1,000); however, a ratio expression of 1:1,000 is equivalent to 1 mg/mL and 1:10,000 is equivalent to 0.1 mg/mL (Ref).

Anaphylaxis and other severe immediate hypersensitivity reactions

Anaphylaxis and other severe immediate hypersensitivity reactions:

Note: IM administration in the anterolateral aspect of the middle third of the thigh is preferred; early administration (including before hospital arrival) is essential to prevent progression to life-threatening manifestations (Ref).

IM: 0.3 or 0.5 mg (use 0.5 mg in patients >50 kg unless patient is rapidly deteriorating, then may use 0.3 mg via autoinjector) using the **1 mg/mL** solution; may repeat every ~5 to 15 minutes (or sooner if clinically indicated) if patient does not adequately respond. Most patients respond to the first or second dose; if inadequate response, additional measures should be instituted (eg, IV fluids and continuous IV epinephrine infusion) (Ref).

IV, intraosseous (alternative routes): Note: In general, reserve IV administration for patients who are unresponsive (eg, profoundly hypotensive) after several epinephrine IM injections. Only use intraosseous when IV access is not readily available; dosing is the same as IV route (Ref).

Continuous infusion (off label): **IV:** Initial: 0.1 to 0.2 mcg/kg/minute (or 8 to 16 mcg/minute for an 80 kg patient) administered with fluid resuscitation; starting dose is dependent on severity of anaphylaxis; titrate every 2 to 3 minutes by 0.05 mcg/kg/minute (or 4 mcg/minute) to response; usual dosing range: 0.01 to 0.2 mcg/kg/minute (or ~1 to 16 mcg/minute for an 80 kg patient) (Ref).

Slow IV bolus (off label): **Note:** Only consider in impending cardiopulmonary arrest when continuous infusion epinephrine is not immediately available; IV bolus administration may be associated with cardiac arrhythmias and cardiac ischemia (Ref).

Initial: 0.05 to 0.1 mg using the **0.1 mg/mL** solution (further diluted in 10 mL of NS) administered over 1 to 10 minutes; may repeat once after 3 minutes if patient remains unresponsive to initial dose. If inadequate response, additional measures should be instituted (eg, IV fluids and continuous IV epinephrine infusion) (Ref). If patient develops cardiopulmonary arrest, use higher IV/intraosseous bolus doses (ie, 1 mg [using the **0.1 mg/mL** solution] every 3 to 5 minutes) administered rapidly (see "Off-label use: Cardiac arrest") (Ref).

Self-treatment using auto-injector or nasal spray for severe allergic reactions (eg, insect stings, food):

Note: For outpatient use, prescribe at least 2 doses for patients who have a prior anaphylactic reaction and for those at risk for severe anaphylaxis (Ref).

Auto-injector: IM (preferred route), SUBQ: 0.3 mg; may repeat every ~5 to 15 minutes (or sooner if clinically indicated) if patient does not adequately respond to initial dose; more than 2 sequential doses should only be administered under direct medical supervision; IM administration is preferred to SUBQ due to more rapid absorption and higher peak concentrations, especially when administered in the thigh (Ref). **Note:** A 0.5 mg autoinjector is available in Canada and may be used or prescribed for patients with more severe symptoms or who weigh >60 kg (Ref).

Nasal spray: Intranasal: One spray (2 mg) into one nostril; if response is inadequate, administer a second dose in the same nostril 5 minutes after the first dose.

Asthma, acute severe

Asthma, acute severe (off-label use):

Note: May use in patients who are unable to use inhaled beta-agonists. Although rare and without formal supportive data, may consider in patients who are refractory to standard therapy and are in impending respiratory arrest (Ref).

IM (preferred route), SUBQ: 0.3 to 0.5 mg using the **1 mg/mL** concentration; if severe symptoms persist after initial dose, may repeat every 20 minutes for a total of 3 doses (Ref). **Note:** IM admin-

istration is preferred to SUBQ due to more rapid absorption and higher peak concentrations when administered in the thigh (Ref).

Atrioventricular block, symptomatic and/or hemodynamically unstable

Atrioventricular block, symptomatic and/or hemodynamically unstable (off-label use):

Note: Should only be used as a temporizing measure until a permanent pacemaker can be placed, if clinically indicated. In most cases, pharmacologic agents will not be effective and temporary transcutaneous or transvenous pacing is necessary as a bridge to permanent pacemaker placement (Ref).

Continuous infusion:

Weight-based dosing: IV: Initial: 0.03 to 0.1 mcg/kg/minute; dosage range: 0.03 to 0.5 mcg/kg/minute; titrate to desired effect (Ref).

Non-weight-based dosing: IV: Initial: 2 to 10 mcg/minute; dosage range: 2 to 40 mcg/minute; titrate to desired effect (doses calculated and rounded for an 80 kg patient based on weight-based dosing using the referenced sources) (Ref).

Bradycardia, symptomatic and/or hemodynamically unstable

Bradycardia, symptomatic and/or hemodynamically unstable (unresponsive to atropine) (off-label use):

Continuous infusion:

Weight-based dosing: IV: Initial: 0.03 to 0.1 mcg/kg/minute; dosage range: 0.03 to 0.5 mcg/kg/minute; titrate to desired effect (Ref).

Non-weight-based dosing: IV: Initial: 2 to 10 mcg/minute; dosage range: 2 to 40 mcg/minute; titrate to desired effect (doses calculated and rounded for an 80 kg patient based on weight-based dosing using the referenced sources) (Ref).

Cardiac arrest

Cardiac arrest (off-label use):

IV, Intraosseous: 1 mg (using the **0.1 mg/mL** solution) every 3 to 5 minutes until return of spontaneous circulation (Ref).

Endotracheal (only use when IV or intraosseous access cannot be established): 2 to 2.5 mg every 3 to 5 minutes until IV/intraosseous access established or return of spontaneous circulation; before administration, dilute in 5 to 10 mL NS or sterile water (using the **1 mg/mL** solution) and use hyperventilation or manual ventilation after administration (Ref). **Note:** May cause false-negative reading with exhaled CO₂ detectors; use second method to confirm tube placement if CO₂ is not detected (Ref).

Hypotension or shock

Hypotension or shock:

Cardiogenic shock (alternative agent) (off-label use):

Note: Typically, not the preferred initial agent in cardiogenic shock; consider other options. Optimal goal of therapy is not well established, but typically titrate to maintain end-organ perfusion (Ref). Institutional protocols may vary with weight-based or non-weight-based dose regimens.

Continuous infusion:

Weight-based dosing:

IV: Usual dosage range: 0.01 to 0.5 mcg/kg/minute; titrate based on clinical end point (eg, BP, end-organ perfusion) (Ref).

Non-weight-based dosing:

IV: Usual dosage range: 1 to 40 mcg/minute; titrate based on clinical end point (eg, BP, end-organ perfusion) (doses calculated and rounded for an 80 kg patient based on weight-based dosing using the referenced sources) (Ref).

Post-cardiac arrest shock (off-label use):

Note: Optimal goal of therapy is not well established, but typically titrate to mean arterial blood pressure (MAP) >65 mm Hg and preferably 80 to 100 mm Hg to optimize cerebral and end-organ perfusion (Ref). Institutional protocols may vary with weight-based or non-weight-based dose regimens.

Continuous infusion:

Weight-based dosing:

IV: Usual dosage range: 0.01 to 1 mcg/**kg**/minute; titrate based on clinical end points (eg, MAP, end-organ perfusion); maximum dose for refractory shock: 2 mcg/**kg**/minute (Ref).

Non-weight-based dosing:

IV: Usual dosage range: 1 to 80 mcg/minute; titrate based on clinical end points (eg, MAP, end-organ perfusion); maximum dose for refractory shock: 160 mcg/minute (doses calculated and rounded for an 80 kg patient based on weight-based dosing using the referenced sources) (Ref).

Septic shock and other vasodilatory shock states (adjunctive or alternative agent):

Note: In patients with shock (eg, sepsis) with cardiac dysfunction who fail to meet resuscitation targets despite adequate fluid status and BP, consider epinephrine monotherapy, particularly in patients with bradyarrhythmia. May also consider in patients requiring additional vasoactive support despite vasopressin and norepinephrine therapy **or** may consider as adjunctive therapy when other vasopressors are not available (Ref). Institutional protocols may vary with weight-based or non-weight-based dose regimens.

Continuous infusion:

Weight-based dosing:

IV: Initial: 0.01 to 0.2 mcg/**kg**/minute; titrate to goal clinical end point (eg, BP or other measure of perfusion); usual dose range: 0.01 to 0.5 mcg/**kg**/minute; maximum dose range for refractory shock: 0.5 to 2 mcg/**kg**/minute (Ref).

Non-weight-based dosing:

IV: Initial: 1 to 15 mcg/minute; titrate to goal clinical endpoint (eg, BP or other measure of perfusion); usual dose range: 1 to 40 mcg/minute; maximum dose range for refractory shock: 40 to 160 mcg/minute (doses calculated and rounded for an 80 kg patient based on weight-based dosing using the referenced sources) (Ref).

Inotropic support

Inotropic support (off-label use):

Note: Inotropic actions predominate at lower doses and vasoconstrictive actions predominate at higher doses (Ref). Institutional protocols may vary with weight-based or non-weight-based dose regimens.

Continuous infusion:

Weight-based dosing:

IV: Usual dosage range: 0.01 to 0.5 mcg/**kg**/minute; titrate based on clinical end point (eg, BP, end-organ perfusion) (Ref).

Non-weight-based dosing:

IV: Usual dosage range: 1 to 40 mcg/minute; titrate based on clinical end point (eg, BP, end-organ perfusion) (doses calculated and rounded for an 80 kg patient based on weight-based dosing using the referenced sources) (Ref).

Mydriasis during intraocular surgery

Mydriasis (induction and maintenance) during intraocular surgery (product specific):

Note: Must **only** use preservative-free vials **without** tartaric acid. Not all formulations of epinephrine injection are suitable for intraocular use; consult the prescribing information for formulation details (Ref). Protocols may vary with concentrations and dose.

Intraocular: Prior to use, must dilute 1 mL of a 1 mg/mL single-use solution to a concentration of 1 to 10 **mcg/mL** or if using for intracameral injection, 2.5 to 10 **mcg/mL**.

Irrigating solution (using 1 to 10 **mcg/mL**): Use as needed during the procedure (Ref).

Intracameral injection (using 2.5 to 10 **mcg/mL**): Injection volume: 0.1 mL (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: Endotracheal, Intranasal, Intraocular, Parenteral: No dosage adjustment likely to be necessary for any degree of kidney impairment (limited excretion in urine as unchanged drug) (Ref).

Hemodialysis, intermittent (thrice weekly): Endotracheal, Intranasal, Intraocular, Parenteral: Not significantly dialyzed (Ref): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: Endotracheal, Intranasal, Intraocular, Parenteral: No dosage adjustment necessary (Ref).

CRRT: Endotracheal, Intranasal, Intraocular, Parenteral: Unlikely to be significantly dialyzable (Ref): No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Endotracheal, Intranasal, Intraocular, Parenteral: No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥ 30 kg/m²):

Continuous infusion: IV: If institution uses weight-based dosing, use **ideal body weight** for initial weight-based dosing calculations, then titrate to hemodynamic effect and clinical response (Ref). If institution uses nonweight-based dosing for vasoactive agents, continue with this approach. During therapy, clinicians should **not** change dosing weight from one weight metric to another (ie, ideal body weight to/from actual body weight or weight-based dosing to/from nonweight-based dosing) (Ref). Refer to adult dosing for indication-specific doses.

Intermittent dosing: IM, SUBQ, intraosseous, endotracheal: Use nonweight-based dosing (Ref). Refer to adult dosing for indication specific doses.

Rationale for recommendations:

There is a paucity of studies evaluating the influence of obesity on epinephrine dosing or pharmacokinetics. Observational studies, including epinephrine, phenylephrine, and norepinephrine, suggest

nonweight-based dosing strategies may result in lower overall cumulative dose requirements and increased drug exposure to second-line agents in some patients and may not be advantageous in time to achieving hemodynamic stability (Ref); however, it is difficult to show outcome differences between weight-based and nonweight-based dosing because of dose titration to target BP, particularly in the context of retrospective studies. Furthermore, there is substantial variability in response in critically ill patients, irrespective of weight. Due to the short onset of action and small V_d , rapid titration to clinical effect after initial dosing is possible (Ref).

In terms of IM administration, there are few pharmacokinetic studies in patients with obesity; however, the data do not demonstrate differences compared to patients without obesity (Ref). An observational, multicenter study in patients presenting with allergic reactions to the emergency room did not find differences in the need for repeat IM dosing between patients with obesity and those without obesity (Ref). Some experts recommend use of a longer 1.5-inch needle to penetrate the subcutaneous fat to maximize effectiveness in patients with obesity, although use of a fixed 0.3 mg dose via autoinjector is recommended in a rapidly deteriorating patient (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Epinephrine (adrenaline) (systemic): Pediatric drug information")

Dosage guidance:

Safety: Epinephrine dosage form concentrations are no longer labeled as ratio expressions (eg, 1:1,000); however, a ratio expression of 1:1,000 is equivalent to 1 mg/mL and 1:10,000 is equivalent to 0.1 mg/mL (Ref). Epinephrine dosing varies by route of administration; use caution.

Asystole or pulseless arrest

Asystole or pulseless arrest (Ref):

Infants, Children, and Adolescents:

IV, Intraosseous (preferred routes of administration): 0.01 mg/kg (0.1 mL/kg of **0.1 mg/mL** solution) immediately after initiating CPR (within 5 minutes of starting chest compressions), maximum dose: 1 mg/dose; repeat every 3 to 5 minutes until return of spontaneous circulation.

Endotracheal: 0.1 mg/kg (0.1 mL/kg of **1 mg/mL** solution) immediately after initiating CPR (within 5 minutes of starting chest compressions), maximum dose: 2.5 mg/dose; repeat every 3 to 5 minutes until return of spontaneous circulation or IV/intraosseous access established. **Note:** Recent clinical studies suggest that lower epinephrine concentrations delivered by endotracheal administration may produce transient beta 2-adrenergic effects which may be detrimental (eg, hypotension, lower coronary artery perfusion pressure). IV or intraosseous are the preferred methods of administration.

Bradycardia

Bradycardia (Ref):

Infants, Children, and Adolescents:

IV, Intraosseous: 0.01 mg/kg (0.1 mL/kg of **0.1 mg/mL** solution) (maximum dose: 1 mg/dose or 10 mL/dose); may repeat every 3 to 5 minutes as needed.

Endotracheal: 0.1 mg/kg (0.1 mL/kg of **1 mg/mL** solution) (maximum dose: 2.5 mg/dose); doses as high as 0.2 mg/kg may be effective; may repeat every 3 to 5 minutes as needed until IV/intraosseous access established. **Note:** Recent clinical studies suggest that lower epinephrine concentrations delivered by endotracheal administration may produce transient beta 2-adrenergic effects which may be detrimental (eg, hypotension, lower coronary artery perfusion pressure). IV or intraosseous are the preferred methods of administration.

Cardiac output increase or maintenance/post resuscitation stabilization

Cardiac output increase or maintenance/post resuscitation stabilization: Infants, Children, and Adolescents: Continuous IV or intraosseous infusion: 0.05 to 1 **mcg/kg/minute**; doses <0.3 **mcg/kg/minute** generally produce beta-adrenergic effects and higher doses (>0.3 **mcg/kg/minute**) generally produce alpha-adrenergic vasoconstriction; titrate dosage to desired effect (Ref).

Hypersensitivity reaction/Anaphylaxis

Hypersensitivity reaction/anaphylaxis:

Note: With parenteral administration, the preferred route of administration is IM administration in the anterolateral aspect of the middle third of the thigh; SUBQ administration results in slower absorption and is less reliable (Ref).

General dosing or health care settings: Infants, Children, and Adolescents: IM: 0.01 mg/kg/dose (0.01 mL/kg/dose of **1 mg/mL** solution) not to exceed: Prepubertal child: 0.3 mg/dose; adolescent: 0.5 mg/dose; administered every 5 to 15 minutes; usual practice is to administer up to 3 doses, but more doses may be needed in some cases (Ref).

Self-/caregiver-administration following type I allergic reactions (eg, insect stings, food):

Autoinjector: Note: If anaphylactic symptoms persist after first dose, may repeat dose in 5 to 15 minutes (Ref).

Manufacturer's labeling (eg, Aarenaclick, Auvi-Q, EpiPen Jr, EpiPen):

Infants, Children, and Adolescents:

7.5 to <15 kg: IM, SUBQ: 0.1 mg; if anaphylactic symptoms persist, dose may be repeated based on severity and response to initial dose; more than 2 sequential doses should only be administered under direct medical supervision.

15 to <30 kg: IM, SUBQ: 0.15 mg; if anaphylactic symptoms persist, dose may be repeated based on severity and response to initial dose; more than 2 sequential doses should only be administered under direct medical supervision.

≥30 kg: IM, SUBQ: 0.3 mg; if anaphylactic symptoms persist, dose may be repeated based on severity and response to initial dose; more than 2 sequential doses should only be administered under direct medical supervision.

Alternate dosing: AAP Recommendations (Ref): Limited data available:

Infants, Children, and Adolescents:

7.5 to <25 kg: IM: 0.15 mg.

≥25 kg: IM: 0.3 mg.

Intranasal : Note: Each Neffy spray is an individual unit (either 1 mg dose or 2 mg dose); a patient should be prescribed or have access to 2 dosing units appropriate for their weight.

Children weighing ≥15 kg and Adolescents:

≥15 kg to <30 kg: Neffy **1 mg/0.1 mL** spray: Intranasal: 1 spray (1 mg) in one nostril once; if symptoms persist or worsen after first dose, may repeat dose in 5 minutes in the same nostril.

≥30 kg: Neffy **2 mg/0.1 mL** spray: Intranasal: 1 spray (2 mg) in one nostril once; if symptoms persist or worsen after first dose, may repeat dose in 5 minutes in the same nostril.

Refractory cases: Infants, Children, and Adolescents: Continuous IV infusion: Prepared 1 **mcg/mL** solution: Initial: 0.1 **mcg/kg/minute**; titrate dose to response. Usual range: 0.1 to 1 **mcg/kg/minute**; maximum dose: 10 **mcg/minute** (Ref). **Note:** Suggested concentration of initial solution is more dilute than those typically utilized in other clinical conditions; evaluate infusion concentration with continued therapy and patient fluid status.

Hypotension/Shock, fluid resistant

Hypotension/Shock, fluid resistant: Infants, Children, and Adolescents: Continuous IV or intraosseous infusion: 0.1 to 1 **mcg/kg/minute**; rates >0.3 **mcg/kg/minute** associated with vasopressor activity (Ref).

Mydriasis during intraocular surgery, induction, and maintenance

Mydriasis during intraocular surgery, induction, and maintenance (product specific):

Note: Not all formulations of epinephrine injection are suitable for intraocular use; consult product labeling prior to use.

Infants, Children, and Adolescents: Intraocular: Must dilute 1 mL of a 1 mg/mL preservative-/sulfite-free, single-use solution to a concentration of 1 **mcg/mL** to 10 **mcg/mL** prior to intraocular use: May use as an irrigation solution as needed during the procedure or may administer intracamerally (ie, directly into the anterior chamber of the eye) with a bolus dose of 0.1 mL of a 2.5 **mcg/mL** to 10 **mcg/mL** dilution.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Injection:

Frequency not defined:

Cardiovascular: Acute myocardial infarction, angina pectoris, cardiac arrhythmia, cardiomyopathy (stress), chest pain, hypertension, increased cardiac work, ischemic heart disease, palpitations, peripheral vasoconstriction, supraventricular tachycardia, tachyarrhythmia, tachycardia, vasoconstriction, ventricular arrhythmia, ventricular ectopy, ventricular fibrillation

Dermatologic: Dermal hemorrhage, diaphoresis, erythema of skin, gangrene of skin and/or subcutaneous tissue (at injection site), pallor, piloerection, skin discoloration

Endocrine & metabolic: Hyperglycemia, hypoglycemia, hypokalemia, insulin resistance, lactic acidosis

Gastrointestinal: Nausea, vomiting

Hematologic & oncologic: Bruise

Local: Localized blanching, tissue necrosis at injection site

Nervous system: Anxiety, apprehension, asthenia, cerebral hemorrhage, cerebrovascular accident, disorientation, dizziness, drowsiness, exacerbation of Parkinson disease, headache, hypoesthesia, memory impairment, panic, paresthesia, psychomotor agitation, restlessness, tingling sensation, tremor

Neuromuscular & skeletal: Limb ischemia

Renal: Kidney impairment

Respiratory: Dyspnea, pulmonary edema, rales

Nasal spray:

Adverse reactions reported in adults unless otherwise indicated.

>10%:

Nervous system: Headache (6% to 18%), jitteriness (children, adolescents, adults: 1% to 11%), paresthesia (children, adolescents: 10%; nose: 19%)

Respiratory: Nasal congestion (children, adolescents: 19%; adults: 2%), nasal discomfort (children, adolescents: 19%; adults: 10% to 13%), respiratory congestion (upper tract: children, adolescents: 14%), rhinorrhea (children, adolescents: 19%; adults: 3% to 7%), sneezing (children, adolescents: 14%; adults: 4%), throat irritation (2% to 19%)

1% to 10%:

Dermatologic: Pruritus of nose (4%)

Gastrointestinal: Abdominal pain (1% to 4%), gingival pain (4%), nausea (2% to 3%), oral hypoesthesia (4%), vomiting (2%)

Nervous system: Dizziness (2% to 3%), fatigue (children, adolescents: 10%), tremor (8%)

Respiratory: Dry nose (children, adolescents: 10%), dry throat (children, adolescents: 10%), epistaxis (children, adolescents: 10%), nasal cavity pain (children, adolescents: 10%)

Contraindications

There are no absolute contraindications to the use of intranasal (Neffy) or injectable epinephrine (including Aarenaclick, Auvi-Q, EpiPen, EpiPen Jr, Symjepi, Allerject [Canadian product], and Twinject [Canadian product]) in a life-threatening situation. The manufacturer's labeling for some injectable products include the following contraindications: Hypersensitivity to sympathomimetic amines; general anesthesia with halogenated hydrocarbons (eg, halothane) or cyclopropane; narrow angle glaucoma; nonanaphylactic shock; in combination with local anesthesia of certain areas such as fingers, toes, and ears; use in situations where vasopressors may be contraindicated (eg, thyrotoxicosis, diabetes, in obstetrics when maternal blood pressure is in excess of 130/80 mm Hg and in hypertension and other cardiovascular disorders).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Injectable solution (Adrenalin, Epinephrine injection, USP): There are no contraindications listed in the manufacturer's labeling.

Warnings/Precautions

Concerns related to adverse effects:

- Cardiac effects: May precipitate or aggravate angina pectoris or induce cardiac arrhythmias; use with caution especially in patients with cardiac disease or those receiving drugs that sensitize the myocardium.
- Extravasation: IV administration: Vesicant; ensure proper needle or catheter placement prior to and during infusion. Avoid extravasation.
- Pulmonary edema: Due to peripheral constriction and cardiac stimulation, pulmonary edema may occur.
- Renal effects: Due to renal blood vessel constriction, decreased urine output may occur.

Disease-related concerns:

- Cardiovascular disease: Use with caution in patients with cardiovascular diseases (eg, arrhythmias, cerebrovascular disease, coronary artery disease, heart disease, hypertension).
- Diabetes: Use with caution in patients with diabetes mellitus; may transiently increase blood glucose levels.
- Hypovolemia: Correct blood volume depletion before administering any vasopressor.

- Parkinson disease: Use with caution in patients with Parkinson disease; psychomotor agitation or temporary worsening of symptoms may occur.
- Pheochromocytoma: Use with caution in patients with pheochromocytoma.
- Thyroid disease: Use with caution in patients with thyroid disease.

Special populations:

- Older adults: Use with caution in older adults.
- Pediatric: Lacerations, bent needles, and embedded needles have been reported in young children who are uncooperative during injection for hypersensitivity reaction using parenteral formulations. To minimize risk, hold the child's leg firmly in place and limit movement prior to and during injection. Although the manufacturers of auto-injectors recommend varying lengths of time for holding the device in the thigh (range: 2 to 10 seconds), longer times have occasionally resulted in injury. For all devices, the needle should remain in the thigh for the least amount of time as possible (~3 seconds) (Brown 2016) (also see "Administration").

Dosage form specific issues:

- Accidental injection: Accidental injection into digits, hands, or feet may result in local reactions including injection-site pallor, coldness, and hypoesthesia or injury resulting in bruising, bleeding, discoloration, erythema, or skeletal injury. Patient should seek immediate medical attention if this occurs.
- Intraocular use: Not all formulations of epinephrine injection are suitable for intraocular use; consult the prescribing information prior to selecting a product. Appropriate products should have an indication for induction and maintenance of mydriasis during intraocular surgery and should not contain any sulfites or preservatives (ISMP 2017). Prior to intraocular use of an appropriate product, must dilute single-use 1 mg/mL (1 mL) solution to a concentration of 1 **mcg**/mL to 10 **mcg**/mL. Corneal endothelial damage has occurred when products containing sodium bisulfite have been used undiluted; therefore, dilution is advised prior to any intraocular use. In addition, products containing chlorobutanol must also not be used intraocularly (may be harmful to corneal endothelium).
- Nasal spray: Absorption may be affected in patients with underlying structural and anatomical nasal conditions (eg, polyps, history of nasal fractures, injuries, or nasal surgery); consider use of injectable epinephrine products in these patients. Nasal mucosa may be altered for up to 2 weeks following intranasal epinephrine administration, which may enhance systemic absorption of intranasally administered medications; risk of adverse effects may be increased.
- Sulfites: Some products may contain sulfites as preservatives. The presence of sulfites in some products should not deter administration during a serious allergic or other emergency situation even if the patient is sulfite-sensitive.

Other warnings/precautions:

- Appropriate use: Hypersensitivity reactions: Do not inject into the buttock; may not effectively treat anaphylaxis and has been associated with Clostridial infections (gas gangrene). Serious skin and soft tissue infections, including necrotizing fasciitis and myonecrosis caused by Clostridia (gas gangrene), have been reported rarely at the injection site. Cleansing skin with alcohol may reduce bacteria at the injection site, but alcohol cleansing does not kill *Clostridium* spores. Preferred injection site is anterolateral aspect of the thigh. Do not administer repeated injections at the same site (tissue necrosis may occur). Monitor for signs/symptoms of injection-site infection.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Kit, Injection:

Epinephrine Professional: 1 mg/mL [contains edetate (edta) disodium dihydrate, sodium metabisulfite]

EPINEPHrinesnap: 1 mg/mL [contains edetate (edta) disodium dihydrate, sodium metabisulfite]

EpinephrineSnap-EMS: 1 mg/mL [contains edetate (edta) disodium dihydrate, sodium metabisulfite]

Epinephrinesnap-v: 1 mg/mL [contains sodium metabisulfite]

EPIsnap: 1 mg/mL [DSC] [contains sodium metabisulfite]

Kit, Injection [preservative free]:

Epinephrinesnap-v: 1 mg/mL [contains edetate (edta) disodium dihydrate, sodium metabisulfite]

Solution, Injection:

Adrenalin: 1 mg/mL (30 mL) [contains chlorobutanol (chlorobutol), disodium edta, sodium metabisulfite]

Generic: 1 mg/mL (1 mL, 10 mL, 30 mL)

Solution, Injection [preservative free]:

Aarenalin: 1 mg/mL (1 mL) [contains edetate (edta) disodium dihydrate, sodium metabisulfite]

Generic: 1 mg/mL (1 mL)

Solution, Intravenous:

Aarenalin: 2 mg/250 mL NaCl 0.9% (250 mL) (250 mL); 10 mg/250 mL NaCl 0.9% (250 mL) (250 mL) [contains edetate (edta) disodium]

Aarenalin: 5 mg/250 mL NaCl 0.9% (250 mL); 8 mg/250 mL in NaCl 0.9% (250 mL) [contains edetate (edta) disodium dihydrate]

Aarenalin-NaCl: 4 mg/250 mL in NaCl 0.9% (250 mL) [contains edetate (edta) disodium]

Solution, Intravenous [preservative free]:

Generic: 16 mg/250 mL NaCl 0.9% (250 mL) (250 mL)

Solution, Nasal:

Neffy: 1 mg/0.1 mL (2 ea); 2 mg/0.1 mL (2 ea) [contains benzalkonium chloride, edetate (edta) disodium, sodium metabisulfite]

Solution Auto-injector, Injection:

Auvi-Q: 0.1 mg/0.1 mL (2 ea); 0.15 mg/0.15 mL (2 ea); 0.3 mg/0.3 mL (2 ea) [contains sodium bisulfite]

EpiPen 2-Pak: 0.3 mg/0.3 mL (1 ea, 2 ea) [contains sodium metabisulfite]

EpiPen Jr 2-Pak: 0.15 mg/0.3 mL (2 ea) [contains sodium metabisulfite]

Generic: 0.15 mg/0.3 mL (1 ea, 2 ea); 0.15 mg/0.15 mL (1 ea, 2 ea); 0.3 mg/0.3 mL (1 ea, 2 ea)

Solution Auto-injector, Injection [preservative free]:

Generic: 0.15 mg/0.3 mL (1 ea, 2 ea)

Solution Prefilled Syringe, Injection [preservative free]:

Symjepi: 0.15 mg/0.3 mL (1 ea [DSC], 2 ea [DSC]); 0.3 mg/0.3 mL (1 ea [DSC], 2 ea [DSC]) [latex free; contains sodium metabisulfite]

Generic: 1 mg/10 mL (10 mL [DSC])

Solution Prefilled Syringe, Intravenous [preservative free]:

Generic: 1 mg/10 mL (10 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Kit (EPINEPHrinesnap Injection)

1 mg/mL (per each): \$148.80

Kit (EpinephrineSnap-EMS Injection)

1 mg/mL (per each): \$104.40

Kit (Epinephrinesnap-v Injection)

1 mg/mL (per each): \$109.20

Solution (Aarenalin Injection)

1 mg/mL (per mL): \$3.92

Solution (Aarenalin Intravenous)

2MG/250ML 0.9% (per mL): \$0.10

5MG/250ML 0.9% (per mL): \$0.24

8MG/250ML 0.9% (per mL): \$0.38

10MG/250ML 0.9% (per mL): \$0.48

Solution (Aarenalin-NaCl Intravenous)

4MG/250ML 0.9% (per mL): \$0.19

Solution (EPINEPHrine Bitartrate-NaCl Intravenous)

16MG/250ML 0.9% (per mL): \$0.70

Solution (EPINEPHrine Injection)

1 mg/mL (per mL): \$7.32 - \$17.95

Solution (EPINEPHrine PF Injection)

1 mg/mL (per mL): \$17.50

Solution (Neffy Nasal)

1MG/0.1ML (per each): \$451.56

2MG/0.1ML (per each): \$451.56

Solution Auto-injector (Auvi-Q Injection)

0.1MG/0.1ML (per each): \$373.15

0.15 mg/0.15 mL (per each): \$373.15

0.3 mg/0.3 mL (per each): \$373.15

Solution Auto-injector (EPINEPHrine Injection)

0.15 mg/0.15 mL (per each): \$247.01

0.15 mg/0.3 mL (per each): \$187.50 - \$346.91

0.3 mg/0.3 mL (per each): \$187.50 - \$346.91

Solution Auto-injector (EpiPen 2-Pak Injection)

0.3 mg/0.3 mL (per each): \$365.17

Solution Auto-injector (EpiPen Jr 2-Pak Injection)

0.15 mg/0.3 mL (per each): \$365.17

Solution Prefilled Syringe (EPINEPHrine Intravenous)

1 mg/10 mL (per mL): \$1.81

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Adrenalin: 1 mg/mL (1 mL, 30 mL) [contains chlorobutanol (chlorobutol), sodium metabisulfite]

Generic: 1 mg/10 mL (10 mL); 1 mg/mL (1 mL, 10 mL)

Solution, Nasal:

Neffy: 2 mg/0.1 mL (1 ea, 2 ea) [contains benzalkonium chloride, edetate (edta) disodium, sodium metabisulfite]

Solution Auto-injector, Intramuscular:

Allerject: 0.15 mg/0.15 mL ([DSC]); 0.3 mg/0.3 mL ([DSC]) [contains sodium metabisulfite]

Anapen: 0.3 mg/0.3 mL ([DSC]) [contains sodium metabisulfite]

Anapen Junior: 0.15 mg/0.3 mL ([DSC]) [contains sodium metabisulfite]

Emerade: 0.3 mg/0.3 mL ([DSC]) [contains disodium edta, sodium metabisulfite]

Emerade: 0.5 mg/0.5 mL ([DSC]) [contains edetate (edta) disodium, sodium metabisulfite]

EpiPen: 0.3 mg/0.3 mL (2 mL) [contains sodium metabisulfite]

EpiPen Jr: 0.15 mg/0.3 mL (2 mL) [contains sodium metabisulfite]

Administration: Adult

Epinephrine solutions for injection can be administered IM, intraosseous, endotracheally, IV, or SUBQ.

Note: Aarenaclick, Allerject [Canadian product], Auvi-Q, EpiPen, EpiPen Jr, and Symjepi contain a single, fixed-dose of epinephrine and may only be administered IM (preferred) or SUBQ. Twinject Auto-Injectors [Canadian product] contain two doses; the first fixed-dose is available for auto-injection; the second dose is available for manual injection following partial disassembly of device.

IV : IV infusions require an infusion pump. When administering as a continuous infusion, central line administration is preferred; extravasation may cause severe ischemic necrosis. If central line is not available, may administer for a short duration (<72 hours) through a peripheral IV catheter placed in a large vein, at a proximal site (eg, in or proximal to antecubital fossa). Frequent monitoring of the IV catheter site is recommended to rapidly identify extravasation (Ref). Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution. Administration through midline catheters may also be an option (Ref).

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution,

then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate phentolamine (or alternative antidote) (Ref).

Phentolamine: **SUBQ:** Dilute 5 to 10 mg in 10 mL 0.9% sodium chloride and administer into extravasation site as soon as possible after extravasation; if infiltrated catheter has not been removed, administer initial dose through the infiltrated catheter; may repeat in 60 minutes if patient remains symptomatic (Ref).

For accidental digital epinephrine injection, inject phentolamine 0.5 to 4.5 mg (diluted in 5 mL 0.9% sodium chloride) divided into multiple SUBQ injections along the area of accidental injury as soon as possible after the injury (Ref).

Alternatives to phentolamine:

Nitroglycerin topical 2% ointment: Apply a 1-inch strip to the site of ischemia to cover the affected areas; may repeat every 8 hours as necessary (Ref).

Terbutaline:

Large extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 1 mg using a solution of terbutaline 1 mg diluted in 10 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

Small extravasations: **SUBQ:** Infiltrate affected extravasation area with 0.5 mg using a solution of terbutaline 1 mg diluted in 1 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

SUBQ: SUBQ administration results in slower absorption and lower peak concentrations (Ref). IM administration is preferred (Ref).

IM: IM administration in the anterolateral aspect of the middle third of the thigh is preferred in the setting of anaphylaxis (Ref). Administer through clothing if necessary. Although the manufacturers of auto-injectors recommend varying lengths of time for holding the device in the thigh (range: 2 to 10 seconds), longer times have occasionally resulted in injury. For all devices, the needle should remain in the thigh for the least amount of time as possible. With nearly every device, the full dose is delivered within 3 seconds. For EpiPen, holding the device against the leg followed by removing the cap and then compressing has been described for easier administration. Never reinsert needles (Ref). Do not administer repeated injections at the same site. Do not inject into the buttocks or into digits, hands, or feet; if injected into the buttocks or into digits, hands, or feet, administer a second dose into the anterolateral aspect of the thigh. Aarenaclick, Aarenalin, Allerject [Canadian product], Auvi-Q, EpiPen, EpiPen Jr, Symjepi, and Twinject [Canadian product] should only be injected into the anterolateral aspect of the thigh.

Obesity: For adults with obesity experiencing anaphylaxis, administer recommended dose using a 1.5-inch needle **or** if patient is rapidly deteriorating, use a fixed-dose autoinjector, if available (Ref). In overweight or obese children, because skin surface to muscle depth is greater in the upper half of the thigh, administration into the lower half of the thigh may be preferred. In very obese children, injection into the calf will provide an even greater chance of intramuscular administration (Ref).

Intranasal: Insert the nozzle of the nasal spray fully into one nostril until fingers touch the nose. Hold the nasal spray straight into the nose; do not angle the nasal spray toward the septum or outer wall of the nose. Use right hand for self-administration into the right nostril or left hand for self-administration into the left nostril. Press the plunger firmly to activate. Do not sniff during or after administration. Neffy product contains a single, fixed dose of epinephrine. Do not prime or reuse for more than one administration. If a second dose is needed, administer into the same nostril via a new nasal spray 5 minutes after the first dose.

Intraocular (product specific): May administer diluted solution (concentration: 1 to 10 mcg/mL) as an irrigation solution as needed during the procedure or may administer intracamerally (ie, directly into the anterior chamber of the eye). **Note:** Must **only** use preservative free vials **without** tartaric acid. Not all formulations of epinephrine injection are suitable for intraocular use; consult the prescribing information. Appropriate products should have mydriasis during intraocular surgery as an approved indication (Ref).

Endotracheal (cardiac arrest when IV or intraosseous access cannot be established): Dilute in NS or sterile water (using the **1 mg/mL** epinephrine solution). Absorption may be greater with sterile water (Ref). Stop compressions, spray drug quickly down tube. Follow immediately with several quick insufflations and continue chest compressions. May cause false-negative reading with exhaled CO₂ detectors; use second method to confirm tube placement if CO₂ is not detected (Ref).

Preparation for Administration: Adult

Endotracheal (off-label route): Dilute in NS or sterile water.

Intraocular: Dilute 1 mL of 1 mg/mL solution in 100 mL to 1,000 mL of an ophthalmic irrigation fluid for a final concentration of 1 to 10 **mcg/mL**; may use this solution as an irrigation as needed during the procedure. May also prepare a dilution of 2.5 to 10 **mcg/mL** for intracameral administration.

Intravenous: Although the manufacturer recommends dilution in dextrose containing solutions (provides protection against significant loss of potency by oxidation) and does not recommend dilution in NS alone, dilution in NS has been reported to be physically compatible (Ref). **Note:** Premixed solutions are available.

Usual Infusion Concentrations: Adult

Note: Premixed solutions are available.

IV infusion: 1 mg in 250 mL (concentration: 4 **mcg/mL**) **or** 4 mg in 250 mL (concentration: 16 **mcg/mL**) of D₅W or NS; 1 mg in 1,000 mL (concentration: 1 **mcg/mL**) in D₅W, D₅NS, or NS.

Administration: Pediatric

Endotracheal:

Neonates: Use **0.1 mg/mL** solution.

Infants, Children, and Adolescents: Use **1 mg/mL** solution; administer and flush with a minimum of 5 mL NS, followed by 5 manual ventilations.

Intranasal: Spray: Neffy: Administration is into a single nostril only; a single Neffy device contains 1 dose; do not prime device.

Insert the nozzle of the nasal spray fully into one nostril until fingers touch the nose. Hold the nasal spray straight into the nose toward the forehead; do not angle the nasal spray toward the septum or outer wall of the nose. Press the plunger firmly to activate. Do not sniff during or after administration. Do not reuse for more than one administration. If a second dose is needed, acquire a new device and administer into the same nostril via a new nasal spray 5 minutes after the first dose.

Intraocular: Administer as a diluted solution (concentration: 1 to 10 **mcg/mL**) as an irrigation solution as needed during the procedure or may administer intracamerally (ie, directly into the anterior chamber of the eye). **Note:** Not all formulations of epinephrine injection are suitable for intraocular use; consult the product labeling prior to use. Appropriate products should have mydriasis during intraocular surgery as an approved indication and should not contain any sulfites or preservatives (Ref).

Parenteral:

Direct IV or intraosseous: If using **0.1 mg/mL** concentration, no dilution is necessary; the **1 mg/mL** concentration must be further diluted.

Continuous IV infusion: Administer as a continuous infusion via an infusion pump. Must be diluted prior to administration. Central line administration is preferred; extravasation may cause severe ischemic necrosis. If central line not available, may administer for a short duration through a peripheral IV catheter placed in a large vein or via intraosseous access using a more dilute solution or with a second carrier fluid (Ref); frequent monitoring of the IV catheter site is recommended to rapidly identify extravasation (Ref). **Note:** Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution. Avoid use of ankle veins (due to potential for gangrene), leg veins in those with occlusive vascular diseases (eg, diabetic endarteritis, Buerger disease, arteriosclerosis, atherosclerosis).

Rate of infusion (mL/hour) = dose (mcg/kg/minute) × weight (kg) × 60 minutes/hour divided by the concentration (mcg/mL)

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses. Initiate phentolamine (or alternative antidote) (Ref).

IM, Autoinjectors (eg, Auvi-Q, EpiPen, EpiPen Jr): Intramuscularly into anterolateral aspect of the middle of the thigh is preferred in the setting of anaphylaxis (Ref). Administer through clothing if necessary. Before administering to a child, immobilize leg to prevent laceration or other injury and consider restraining children who are likely to move during administration. Although the manufacturers of auto-injectors recommend varying lengths of time for holding the device in the thigh (range: 2 to 10 seconds), longer times have occasionally resulted in injury. For all devices, the needle should remain in the thigh for the least amount of time as possible. With nearly every device, the full dose is delivered within 3 seconds. For EpiPen brand, performing in a different order may lessen risk of injury to hand: remove device from protective case, hold the device against the leg, then remove the safety cap, and then compress. Never reinsert needles (Ref). Do not administer repeated injections at the same site. Do not inject into the buttocks or into digits, hands, or feet. If injected into buttocks, administer a second dose into the anterolateral aspect of the thigh. If injected into the digits, hands, or feet, administer a second dose into the anterolateral aspect of the thigh and immediately go to the nearest emergency room (inform health care provider of location of accidental injection). More than 2 sequential doses should only be administered under direct medical supervision. Some products are single-use products and a new device should be used for each dose; consult product-specific labeling. **Note:** In children who are overweight or obese, because skin surface to muscle depth is greater in the upper half of the thigh, administration into the lower half of the thigh may be preferred. In children who are very obese, injection into the calf will provide an even greater chance of intramuscular administration (Ref).

SUBQ: Not preferred; IM administration achieves peak concentrations faster (Ref).

Autoinjectors: Autoinjectors may be administered subcutaneously into the anterolateral aspect of middle of the thigh (Ref); however, due to the needle length, it is likely that it will be administered IM except in adolescents with obesity (Ref).

1 mg/mL solution: Administer into the anterolateral aspect of the thigh (Ref).

Preparation for Administration: Pediatric

Endotracheal: May dilute in 1 to 5 mL NS based on patient size (Ref).

Intraocular: **Note:** Product specific; not all formulations of epinephrine injection are suitable for intraocular use; consult the product labeling prior to use: Dilute 1 mL of **1 mg/mL** preservative-/sulfite-free single-use solution in 100 mL to 1,000 mL of an ophthalmic irrigation fluid for a final concentration of 1 to 10 **mcg/mL**; may use this solution as an irrigation as needed during the procedure. May also prepare a dilution of 2.5 to 10 **mcg/mL** for intracameral administration. Do not use 30 mL vial for preparation.

Parenteral:

Direct IV or intraosseous administration: Dilute to a maximum concentration of 100 mcg/mL (0.1 mg/mL); if using **0.1 mg/mL** concentration, no dilution is necessary.

Continuous IV infusion: Concentration may vary by use.

Shock/hypotension; cardiac output maintenance/stabilization, postresuscitation: Dilute to a maximum concentration of 64 mcg/mL (Ref). Vermont Oxford Network recommends standard concentrations of 5 mcg/mL and 10 mcg/mL for neonates (Ref). For pediatric patients, ASHP recommends standard concentrations of 10 mcg/mL, 20 mcg/mL, and 40 mcg/mL (Ref). **Note:** Although the manufacturer recommends dilution in dextrose-containing solutions (provides protection against significant loss of potency by oxidation) and does not recommend dilution in NS alone, dilution in NS has been reported to be physically compatible (Ref).

Hypersensitivity/anaphylaxis (refractory): Prepare a 1 mcg/mL solution (eg, add 1 mg of the **1 mg/mL** solution to 1,000 mL of D5W or NS) (Ref).

Usual Infusion Concentrations: Pediatric

IV infusion: 10 **mcg/mL**, 20 **mcg/mL**, 32 **mcg/mL**, 40 **mcg/mL**, or 64 **mcg/mL**.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Epinephrine is sensitive to light and air. Protection from light is recommended. Oxidation turns drug pink, then a brown color. **Solutions should not be used if they are discolored or contain a precipitate.**

Arenaclick: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F); do not freeze or refrigerate.

Adrenalin:

Premixed solution: Store at 20°C to 25°C (68°F to 77°F). Protect from freezing. Keep in foil overwrap until ready to use. Discard after 24 hours of opening overwrap.

Vials: Store intact vials at 20°C to 25°C (68°F to 77°F); do not freeze. Discard multiple-dose vials 30 days after initial use. Diluted solution for infusion may be stored for up to 4 hours at room temperature or for 24 hours refrigerated.

Allerject [Canadian product]: Store at 15°C to 30°C (59°F to 86°F); do not refrigerate.

Auvi-Q: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F); do not freeze.

EpiPen, EpiPen Jr, and Symjepi: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F); do not freeze or refrigerate.

Neffy: Store at 20°C to 25°C (68°F to 77°F); excursions permitted up to 50°C (122°F). Do not freeze; however, it may be used if thawed after being previously frozen. Keep in blister pack or carry case until ready to use.

Premixed solution (Viaflo container): Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Keep in overpouch until ready to use.

Twinject [Canadian product]: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F); do not freeze or refrigerate. Protect from light.

Vials for injection: Store intact vials at 20°C to 25°C (68°F to 77°F); do not refrigerate; protect from freezing and from alkalis and oxidizing agents. Discard multiple-dose vials 30 days after initial use. Diluted solution for infusion may be stored for up to 4 hours at room temperature or for 24 hours refrigerated (Fresenius Kabi prescribing information).

Use: Labeled Indications

Anaphylaxis and other severe immediate hypersensitivity reactions: Treatment of type I allergic reactions, including anaphylactic reactions.

Hypotension or shock: Treatment of hypotension associated with septic shock in adults (to increase mean arterial BP).

Mydriasis during intraocular surgery (product specific): Induction and maintenance of mydriasis during intraocular surgery. **Note:** Not all formulations of epinephrine injection are suitable for intraocular use; consult the prescribing information.

Use: Off-Label: Adult

Asthma, acute severe; Atrioventricular block, symptomatic and/or hemodynamically unstable; Bradycardia, symptomatic and/or hemodynamically unstable (unresponsive to atropine); Cardiac arrest; Cardiogenic shock; Inotropic support; Post-cardiac arrest shock

Medication Safety Issues

Sound-alike/look-alike issues:

Arenalin may be confused with Akovaz

EPINEPHrine may be confused with ePHEDrine

Epifrin may be confused with ephedrine, EpiPen

Neffy (nasal spray) may be confused with nalaxone (nasal sprays [Kloxxado, Narcan, Rextovy, Re-Vive])

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication (IM, SUBQ administration) and also this drug class (adrenergic agonist, IV) among its list of drugs/classes which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

Administration issues:

Medication errors have occurred due to confusion with epinephrine products expressed as ratio strengths (eg, 1:1000 vs 1:10,000).

Epinephrine 1:1000 = 1 mg/mL and is most commonly used IM

Epinephrine 1:10,000 = 0.1 mg/mL and is used IV

Medication errors have occurred when topical epinephrine 1 mg/mL (1:1000) has been inadvertently injected. Vials of injectable and topical epinephrine look very similar. Epinephrine should always be appropriately labeled with the intended administration.

EpiPen 0.3 mg and EpiPen Jr 0.15 mg auto-injectors, and authorized generic versions, may potentially have delayed injection or be prevented from properly injecting due to device failure from spontaneous activation caused by using sideways force to remove the blue safety release, device failure from inadvertent or spontaneous activation due to a raised blue safety release, difficulty removing the device from the carrier tube, and user errors. Health care providers, patients, and caregivers should periodically review the EpiPen user instructions and practice using the EpiPen trainer to ensure proper understanding and utilization of the EpiPen auto-injector. Patients and caregivers should inspect their epinephrine autoinjector prior to needing it to ensure the blue safety release is not raised and that the device can be easily removed from the carrier tube (FDA 2020).

International issues:

EpiPen [US, Canada, and multiple international markets] may be confused with Epigen brand name for glycyrrhizinic acid [Argentina, Mexico, Russia] and Epopen brand name for epoetin alfa [Spain] and Epigent brand name for gentamicin [multiple international markets]

Metabolism/Transport Effects

Substrate of COMT

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug

interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alpha1-Blockers: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Atomoxetine: May increase hypertensive effects of Sympathomimetics. Atomoxetine may increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Azosemide: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Benperidol: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Benzylpenicilloyl Polylysine: Coadministration of Alpha-/Beta-Agonists and Benzylpenicilloyl Polylysine may alter diagnostic results. Management: Consider delaying skin testing until alpha/beta agonists are no longer required and their effects have diminished, or use a histamine skin test as a positive control to assess a patient’s ability to mount a wheal and flare response. *Risk D: Consider Therapy Modification*

Beta-Blockers (Beta1 Selective): May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Beta-Blockers (Nonselective): May increase hypertensive effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Beta-Blockers (with Alpha-Blocking Properties): May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Bitter Orange: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Blonanserin: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk X: Avoid*

Bornaprine: Sympathomimetics may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Bromocriptine: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives to this combination when possible. If combined, monitor for hypertension and tachycardia, and do not coadminister these agents for more than 10 days. *Risk D: Consider Therapy Modification*

Bromperidol: May decrease therapeutic effects of EPINEPHrine (Systemic). *Risk X: Avoid*

Cannabinoid-Containing Products: May increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Cardiac Glycosides: EPINEPHrine (Systemic) may increase arrhythmogenic effects of Cardiac Glycosides. *Risk C: Monitor*

Chlorprocaine (Systemic): May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Chlorprothixene: May increase adverse/toxic effects of EPINEPHrine (Systemic). Specifically, paradoxical hypotension and tachycardia may be increased. EPINEPHrine (Systemic) may increase therapeutic effects of Chlorprothixene. *Risk C: Monitor*

CloZAPine: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Cocaine (Topical): May increase hypertensive effects of Sympathomimetics. Management: Consider alternatives to use of this combination when possible. Monitor closely for substantially increased blood pressure or heart rate and for any evidence of myocardial ischemia with concurrent use. *Risk D: Consider Therapy Modification*

COMT Inhibitors: May increase serum concentrations of COMT Substrates. *Risk C: Monitor*

Dihydralazine: Sympathomimetics may decrease therapeutic effects of Dihydralazine. *Risk C: Monitor*

Diuretics: May increase arrhythmogenic effects of EPINEPHrine (Systemic). Diuretics may decrease vasopressor effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Doxapram: Sympathomimetics may increase adverse/toxic effects of Doxapram. *Risk C: Monitor*

Doxofylline: Sympathomimetics may increase adverse/toxic effects of Doxofylline. *Risk C: Monitor*

Droxidopa: Sympathomimetics may increase hypertensive effects of Droxidopa. *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): May increase vasoconstricting effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Esketamine (Injection): May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

Guanethidine: May increase hypertensive effects of Sympathomimetics. Guanethidine may increase arrhythmogenic effects of Sympathomimetics. *Risk C: Monitor*

Haloperidol: May decrease vasoconstricting effects of EPINEPHrine (Systemic). Management: Consider alternatives to this combination and monitor for reduced epinephrine efficacy, and possible paradoxical effects (ie, hypotension), when combined. Use of alternative vasopressor agents (eg, phenylephrine, metaraminol, norepinephrine) is preferred. *Risk D: Consider Therapy Modification*

Hexoprenaline: May increase adverse/toxic effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Hyaluronidase: May increase vasoconstricting effects of Alpha-/Beta-Agonists. Management: Do not use hyaluronidase to enhance the dispersion or absorption of alpha-/beta-agonists. Use of hyaluronidase for other purposes in patients receiving alpha-/beta-agonists may be considered as clinically indicated. *Risk D: Consider Therapy Modification*

Inhalational Anesthetics: May increase arrhythmogenic effects of EPINEPHrine (Systemic). Management: Administer epinephrine with added caution in patients receiving, or who have recently received, inhalational anesthetics. Use lower than normal doses of epinephrine and monitor for the development of cardiac arrhythmias. *Risk D: Consider Therapy Modification*

Isoproterenol: May increase therapeutic effects of EPINEPHrine (Systemic). *Risk X: Avoid*

Kratom: May increase adverse/toxic effects of Sympathomimetics. *Risk X: Avoid*

Linezolid: May increase hypertensive effects of Sympathomimetics. Management: Consider initial dose reductions of sympathomimetic agents, and closely monitor for enhanced blood pressure elevations, in patients receiving linezolid. *Risk D: Consider Therapy Modification*

Lisuride: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives when possible, as product labeling states the concomitant use of lisuride with alpha-/beta-agonists is not recommended. If combined, monitor for increases in blood pressure or hypertensive crises. *Risk D: Consider Therapy Modification*

Melperone: May increase adverse/toxic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase hypertensive effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Pergolide: May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Promethazine: May decrease vasoconstricting effects of EPINEPHrine (Systemic). Management: Avoid epinephrine and consider norepinephrine or phenylephrine when treating hypotension due to promethazine overdose. Consider alternative vasoconstrictors in patients treated with promethazine. This combination may be indicated in anaphylaxis treatment. *Risk D: Consider Therapy Modification*

Serotonin/Norepinephrine Reuptake Inhibitor: May increase tachycardic effects of Alpha-/Beta-Agonists. Serotonin/Norepinephrine Reuptake Inhibitor may increase vasopressor effects of Alpha-/Beta-Agonists. Management: If possible, avoid coadministration of direct-acting alpha-/beta-agonists and serotonin/norepinephrine reuptake inhibitors. If coadministered, monitor for increased sympathomimetic effects (eg, increased blood pressure, chest pain, headache). *Risk D: Consider Therapy Modification*

Solriamfetol: Sympathomimetics may increase hypertensive effects of Solriamfetol. Sympathomimetics may increase tachycardic effects of Solriamfetol. *Risk C: Monitor*

Sympathomimetics: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Tedizolid: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for increased blood pressure and heart rate may be increased. *Risk C: Monitor*

Thyroid Products: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk of hypertension, tachycardia, and coronary insufficiency may be increased in patients with coronary artery disease. *Risk C: Monitor*

Tricyclic Antidepressants: May increase vasopressor effects of Alpha-/Beta-Agonists. Management: Avoid, if possible, the use of alpha-/beta-agonists in patients receiving tricyclic antidepressants. If combined, monitor for evidence of increased pressor effects and consider reductions in initial dosages of the alpha-/beta-agonist. *Risk D: Consider Therapy Modification*

Vasopressin: Alpha-/Beta-Agonists (Direct-Acting) may increase hypertensive effects of Vasopressin. The effect of other hemodynamic parameters may also be enhanced. *Risk C: Monitor*

Pregnancy Considerations

Epinephrine crosses the placenta (Sandler 1964).

Epinephrine is recommended for the treatment of anaphylaxis in pregnant women. Specific dosing is not available; use with caution and monitor hemodynamic response (Hepner 2013). Medications used for the treatment of cardiac arrest in pregnancy are the same as in the non-pregnant woman. Doses and indications should follow current Advanced Cardiovascular Life Support guidelines. Appropriate medications should not be withheld due to concerns of fetal teratogenicity (AHA [Jeejeebhoy 2015]).

Breastfeeding Considerations

It is not known if epinephrine is present in breast milk.

Epinephrine is generally considered compatible in breastfeeding and is recommended for the treatment of anaphylaxis in breastfeeding women (WHO 2002).

Monitoring Parameters

Heart rate, blood pressure (invasive blood pressure monitoring recommended while receiving continuous infusion); monitor site of infusion for blanching/extravasation; continuous cardiac monitoring required during continuous infusion. If using to treat hypotension, assess intravascular volume prior to and during therapy; support as needed; monitor for cardiac arrhythmias, hyperlactatemia, and hyperglycemia.

Consult individual institutional policies and procedures.

Mechanism of Action

Stimulates alpha-, beta₁-, and beta₂-adrenergic receptors resulting in relaxation of smooth muscle of the bronchial tree, cardiac stimulation (increasing myocardial oxygen consumption), and dilation of skeletal muscle vasculature; small doses can cause vasodilation via beta₂-vascular receptors; large doses may produce constriction of skeletal and vascular smooth muscle

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Bronchodilation: SUBQ: ~5 to 10 minutes

Absorption: Orally ingested doses are rapidly metabolized in GI tract and liver; pharmacologically active concentrations are not achieved

Distribution: Does not cross blood-brain barrier

Metabolism: Taken up into the adrenergic neuron and metabolized by monoamine oxidase and catechol-o-methyltransferase; circulating drug hepatically metabolized

Half-life elimination: IV: <5 minutes

Excretion: Urine (as inactive metabolites, metanephrine, and sulfate and hydroxy derivatives of mandelic acid, small amounts as unchanged drug)

Patient Counseling Points

(For additional information see "Epinephrine (adrenaline) (systemic): Patient drug information")

What is this drug used for?

- It is used to treat an allergic reaction.
- It is used to treat low blood pressure.
- It is used during eye surgery.
- It is used when the heart is not beating.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Feeling anxious, nervous, excitable, shaky, or restless
- Upset stomach or throwing up
- Sweating a lot
- Pale skin
- Headache

Injection:

- Feeling dizzy, tired, or weak
- Goosebumps

Nose spray:

- Nose or throat irritation
- Nosebleed
- Tingling
- Feeling jittery
- Runny or stuffy nose
- Dry nose or throat
- Sneezing
- Feeling dizzy, sleepy, tired, or weak
- Stomach pain
- Numbness in the mouth
- Pain in gums of teeth

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- High blood pressure like very bad headache or dizziness, passing out, or change in eyesight

- Chest pain or pressure
- A fast heartbeat
- Shortness of breath
- Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight
- Abnormal heartbeat
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

All injection products:

- Low potassium levels like muscle pain or weakness, muscle cramps, or a heartbeat that does not feel normal
- Not able to pass urine or change in how much urine is passed
- A burning, numbness, or tingling feeling that is not normal

Injection (for allergic reactions):

- Infection like redness that does not go away, warmth, swelling, or tenderness

Injection (if given in the vein):

- Redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body

Nose spray:

- Feeling confused
- Memory problems or loss

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Arenaline | Epipen | Jext;
- (AR) Argentina: Adrenalina Bioquim | Adrenalina drawer | Epipen | Epipen jr;
- (AT) Austria: Arenaline aguettant | Emerade | Epipen | L-aarenalin;
- (AU) Australia: Arenaline | Arenaline aguettant | Arenaline jr mylan | Arenaline mylan | Emerade | Epipen;

- (BE) Belgium: Aarenaline (tartrate) aguettant | Aarenaline Parke-Davis | Anahelp | Epipen;
- (BG) Bulgaria: Aarenalin;
- (CH) Switzerland: Aarenalin aguettant | Adrenalin labatec | Emerade | Epipen;
- (CL) Chile: Epipen;
- (CO) Colombia: Epinefrina | Penepin jr;
- (CZ) Czech Republic: Aarenalin | Emeraæ | Epipen;
- (DE) Germany: ADRENALIN Carino | Aarenalin ethypharm | Aarenalin Infectopharm | Anapen | Emerade | Epipen | Fastjekt | Fastjekt junior | Jext | Min-I-Jet Aarenalin;
- (DK) Denmark: Epipen;
- (DO) Dominican Republic: Adrenalina;
- (EE) Estonia: Aarenalin aguettant | Adrenalini hcl | Epipen;
- (EG) Egypt: Aarenaline;
- (ES) Spain: Adreject | Adrenalina level | Altellus | Anapen | Emeraæ;
- (FI) Finland: Aarenalin aguettant | Aarenalin stragen | Emeraæ | Epipen;
- (FR) France: Aarenaline aguettant | Aarenaline b.braun | Aarenaline cooper | Aarenaline PCH | Anahelp | Anakit | Anapen | Emeraæ | Epipen;
- (GB) United Kingdom: Aarenaline | Emerade | Epipen | Ims aarenaline | Jext;
- (GR) Greece: Adrenalina wzf | Anapen | Emeraæ | Epipen | Fastjekt | Jext;
- (HK) Hong Kong: Aarenaline aguettant | Epinephrine minijet;
- (HR) Croatia: Epipen | Epipen jr;
- (ID) Indonesia: Aarenaline | Adrenalini bitartras;
- (IE) Ireland: Aarenaline | Anapen | Epipen;
- (IL) Israel: Epipen;
- (IT) Italy: Adrenalina | Adrenalina Aguettant | Chenpen | Fastjekt | Fastjekt junior;
- (JP) Japan: Epipen | Epiquick;
- (KE) Kenya: Aarenaline aguettant;
- (KR) Korea, Republic of: Jext;
- (KW) Kuwait: Aarenaline | Aarenaline aguettant | Epipen | Epipen jr;
- (LT) Lithuania: Adrenalin | Adrenalina wzf | Fastjekt | Jext;
- (LU) Luxembourg: Aarenaline (tartrate) aguettant | Fastjekt | Jext;
- (LV) Latvia: Aarenalin | Adrenalini | Adrenalini hydrotartras;
- (MY) Malaysia: Aarenaline;
- (NL) Netherlands: Aarenaline | Emerade | Epipen | Jext;
- (NO) Norway: Adrenalin aguettant | Aarenalin stragen | Aarenaline aurum | Emeraæ | Epipen;
- (NZ) New Zealand: Aarenaline aguettant | Epipen;
- (PA) Panama: Adrenalina Aguettant;
- (PK) Pakistan: Aarenaline;
- (PL) Poland: Adrenalina Aguettant | Adrenalina wzf | Adrenalinum | Fastjekt;
- (PR) Puerto Rico: Adrenalin chloride | Epinephrine | Epipen | Epipen 2-Pak | Epipen jr 2-pak | Symjepi;
- (PT) Portugal: Adrenalina | Emeraæ | Epipen | Spinefe;
- (PY) Paraguay: Adrenalina lasca;
- (QA) Qatar: Emerade | EpiPen | Jext | Penepin | Penepin Jr.;
- (RO) Romania: Anapen | Epipen;
- (RU) Russian Federation: Aarenalin | Epigect;
- (SA) Saudi Arabia: Aarenaline | Epipen | Jext;
- (SE) Sweden: Adrenalin mylan | Aarenalin stragen | Anapen | Emeraæ | Epipen | Epipen jr | Jext;
- (SG) Singapore: Aarenaline;
- (SI) Slovenia: Aarenalin aguettant | Epipen;
- (SK) Slovakia: Aarenalin | Epipen;
- (SR) Suriname: Anapen | Epipen | Jext;
- (TN) Tunisia: Anapen;
- (TR) Turkey: Aarenalin | Penepin;
- (TW) Taiwan: Bosmin | Epirenamin;
- (UA) Ukraine: Aarenalin | Epipen | Epipen junior;
- (UG) Uganda: Aarenaline aguettant;

- (UY) Uruguay: Adrenalina;
- (VE) Venezuela, Bolivarian Republic of: Adrenalina;
- (ZA) South Africa: Aarenaline | Adrenotone | Anaguard | EpiPen

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